

# Self Study Assignment 3

Submission Deadline: 6th Jan(Sec-5) 7th | Jan(Sec-07)

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Consider the following SDD and evaluate the following string:

```

if (a > b || b < c && c > d)
    if (a > c || c > d)
        b = c + d + e;
        e = d + c;
        while (c > e)
            c = c + f;
    else
        e = e + f + g;

```

SDD:

| PRODUCTION                | SEMANTIC RULES                                                                                                       |
|---------------------------|----------------------------------------------------------------------------------------------------------------------|
| $P \rightarrow S$         | $S.next = newlabel()$<br>$P.code = S.code \parallel label(S.next)$                                                   |
| $S \rightarrow id = E ;$  | $S.code = E.code \parallel$<br>$gen(top.get(id.lexeme) '=' E.addr)$                                                  |
| $E \rightarrow E_1 + E_2$ | $E.addr = new Temp()$<br>$E.code = E_1.code \parallel E_2.code \parallel$<br>$gen(E.addr '=' E_1.addr '+' E_2.addr)$ |
| $  - E_1$                 | $E.addr = new Temp()$<br>$E.code = E_1.code \parallel$<br>$gen(E.addr '=' 'minus' E_1.addr)$                         |
| $  ( E_1 )$               | $E.addr = E_1.addr$<br>$E.code = E_1.code$                                                                           |
| $  id$                    | $E.addr = top.get(id.lexeme)$<br>$E.code = ''$                                                                       |

|                                                       |                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $S \rightarrow \text{if} ( B ) S_1$                   | $B.true = \text{newlabel}()$<br>$B.false = S_1.next = S.next$<br>$S.code = B.code \parallel \text{label}(B.true) \parallel S_1.code$                                                                                                                                                            |
| $S \rightarrow \text{if} ( B ) S_1 \text{ else } S_2$ | $B.true = \text{newlabel}()$<br>$B.false = \text{newlabel}()$<br>$S_1.next = S_2.next = S.next$<br>$S.code = B.code$<br>$\quad \parallel \text{label}(B.true) \parallel S_1.code$<br>$\quad \parallel \text{gen('goto' } S.next)$<br>$\quad \parallel \text{label}(B.false) \parallel S_2.code$ |
| $S \rightarrow \text{while} ( B ) S_1$                | $begin = \text{newlabel}()$<br>$B.true = \text{newlabel}()$<br>$B.false = S.next$<br>$S_1.next = begin$<br>$S.code = \text{label}(begin) \parallel B.code$<br>$\quad \parallel \text{label}(B.true) \parallel S_1.code$<br>$\quad \parallel \text{gen('goto' } begin)$                          |
| $S \rightarrow S_1 S_2$                               | $S_1.next = \text{newlabel}()$<br>$S_2.next = S.next$<br>$S.code = S_1.code \parallel \text{label}(S_1.next) \parallel S_2.code$                                                                                                                                                                |

|                                      |                                                                                                                                                                      |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $B \rightarrow B_1 \parallel B_2$    | $B_1.true = B.true$<br>$B_1.false = newlabel()$<br>$B_2.true = B.true$<br>$B_2.false = B.false$<br>$B.code = B_1.code \parallel label(B_1.false) \parallel B_2.code$ |
| $B \rightarrow B_1 \&\& B_2$         | $B_1.true = newlabel()$<br>$B_1.false = B.false$<br>$B_2.true = B.true$<br>$B_2.false = B.false$<br>$B.code = B_1.code \parallel label(B_1.true) \parallel B_2.code$ |
| $B \rightarrow ! B_1$                | $B_1.true = B.false$<br>$B_1.false = B.true$<br>$B.code = B_1.code$                                                                                                  |
| $B \rightarrow E_1 \text{ rel } E_2$ | $B.code = E_1.code \parallel E_2.code$<br>$\parallel gen('if' E_1.addr \text{ rel.op } E_2.addr 'goto' B.true)$<br>$\parallel gen('goto' B.false)$                   |
| $B \rightarrow \text{true}$          | $B.code = gen('goto' B.true)$                                                                                                                                        |
| $B \rightarrow \text{false}$         | $B.code = gen('goto' B.false)$                                                                                                                                       |

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## The whole SDD:

| PRODUCTION                            | SEMANTIC RULES                                                                                                                                                                                                                            |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $P \rightarrow S$                     | $S.next = newlabel()$<br>$P.code = S.code \parallel label(S.next)$                                                                                                                                                                        |
| $S \rightarrow id = E ;$              | $S.code = E.code \parallel$<br>$gen(top.get(id.lexeme) ' = ' E.addr)$                                                                                                                                                                     |
| $E \rightarrow E_1 + E_2$             | $E.addr = new Temp()$<br>$E.code = E_1.code \parallel E_2.code \parallel$<br>$gen(E.addr ' = ' E_1.addr ' + ' E_2.addr)$                                                                                                                  |
| $\quad   - E_1$                       | $E.addr = new Temp()$<br>$E.code = E_1.code \parallel$<br>$gen(E.addr ' = ' minus ' E_1.addr)$                                                                                                                                            |
| $\quad   ( E_1 )$                     | $E.addr = E_1.addr$<br>$E.code = E_1.code$                                                                                                                                                                                                |
| $\quad   id$                          | $E.addr = top.get(id.lexeme)$<br>$E.code = ''$                                                                                                                                                                                            |
| $S \rightarrow if ( B ) S_1$          | $B.true = newlabel()$<br>$B.false = S_1.next = S.next$<br>$S.code = B.code \parallel label(B.true) \parallel S_1.code$                                                                                                                    |
| $S \rightarrow if ( B ) S_1 else S_2$ | $B.true = newlabel()$<br>$B.false = newlabel()$<br>$S_1.next = S_2.next = S.next$<br>$S.code = B.code$<br>$\parallel label(B.true) \parallel S_1.code$<br>$\parallel gen('goto' S.next)$<br>$\parallel label(B.false) \parallel S_2.code$ |
| $S \rightarrow while ( B ) S_1$       | $begin = newlabel()$<br>$B.true = newlabel()$<br>$B.false = S.next$<br>$S_1.next = begin$<br>$S.code = label(begin) \parallel B.code$<br>$\parallel label(B.true) \parallel S_1.code$<br>$\parallel gen('goto' begin)$                    |
| $S \rightarrow S_1 S_2$               | $S_1.next = newlabel()$<br>$S_2.next = S.next$<br>$S.code = S_1.code \parallel label(S_1.next) \parallel S_2.code$                                                                                                                        |
| $B \rightarrow B_1    B_2$            | $B_1.true = B.true$<br>$B_1.false = newlabel()$<br>$B_2.true = B.true$<br>$B_2.false = B.false$<br>$B.code = B_1.code \parallel label(B_1.false) \parallel B_2.code$                                                                      |
| $B \rightarrow B_1 \&\& B_2$          | $B_1.true = newlabel()$<br>$B_1.false = B.false$<br>$B_2.true = B.true$<br>$B_2.false = B.false$<br>$B.code = B_1.code \parallel label(B_1.true) \parallel B_2.code$                                                                      |
| $B \rightarrow ! B_1$                 | $B_1.true = B.false$<br>$B_1.false = B.true$<br>$B.code = B_1.code$                                                                                                                                                                       |
| $B \rightarrow E_1 rel E_2$           | $B.code = E_1.code \parallel E_2.code$<br>$\parallel gen('if' E_1.addr rel.op E_2.addr 'goto' B.true)$<br>$\parallel gen('goto' B.false)$                                                                                                 |
| $B \rightarrow true$                  | $B.code = gen('goto' B.true)$                                                                                                                                                                                                             |
| $B \rightarrow false$                 | $B.code = gen('goto' B.false)$                                                                                                                                                                                                            |